

Designing a flexible tool for rapid implementation of brain-computer interfaces (BCI) in game development

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Abstract— Early neurological injury or disease can lead to severe life-long physical impairments, despite normal cognitive function. For such individuals, brain-computer interfaces (BCI) may provide a means to regain access to the world by offering control of systems through directly processing brain patterns. However, current BCI applications are often research driven and consequently seen as uninteresting, particularly for prolonged use and younger BCI-users. To help mitigate this concern, this paper establishes a tool for researchers and game developers alike to rapidly incorporate a BCI control scheme (the P300 oddball response) into a gaming environment. Preliminary results indicate the proposed P300 Dynamic Cube (PDC) asset works in online BCI environments ($n=20$, healthy adult participants), resulting in median classification accuracy of $75 \pm 3.28\%$. Additionally, the PDC tool can be rapidly adapted for a variety of game designs, evidenced by its incorporation into submissions to the Brain-Computer Interface (BCI) Game Jam 2019 competition. These findings support the PDC as a useful asset in the design and development of BCI-based games.

Clinical Relevance— This paper establishes an open-source, flexible, and accessible tool designed to ease development and incorporation of BCI control schemes into video games based on the Unity Game Engine.

I. INTRODUCTION

Profound physical disability is an everyday challenge for many individuals, and a life-long struggle for those who have suffered early-life neurological disease or injury. Despite being cognitively normal, physical limitations can bar these individuals from interacting with the world. Brain-computer interfaces (BCI) are an emerging technology which offer such users a way to influence their environment by directly processing their brain patterns and bypassing potential neuromuscular limitations [1], [2].

BCI technology is still in early stages, but is rapidly gaining popularity [2]. Current BCI paradigms often focus on simple, utility-driven applications for adults such as spelling or operating simple switches. While useful, these applications may have limited appeal for sustained use, particularly for young BCI-users. Despite their severe physical disability, many such children remain intellectually curious and express typical interests in music, art and entertainment. Evidence indicates that increasing engagement in BCI through gamified learning can result in longer adoption of the technology and help promote practicing BCI control schemes [3]–[8]. Thus while BCI gaming may be particularly motivating for young

BCI users, it has potential to improve quality of life for the wider BCI population.

BCI game applications are often impeded in their design, due in part to the games' development environment. Literature reveals many BCI games are designed and built in-house by research teams in order to address their specific research questions [5], [7], [9]–[11]. While this is reasonable from a human resources point of view, it can lead to independent researchers continuously recreating very similar tools which incorporate BCI control schemes into a gaming environment like the Unity Game Engine (<https://unity.com/>). Additionally, limited practical details are provided on how the BCI input is actually integrated into the game environment. As an example, work reported by Rohani et al. in 2014 [3] offers minimal insight into the implementation of BCI in Unity software. In other scenarios, BCI games are tied directly to commercial hardware, which also significantly limits design as they cannot be extensively customized for individual interests.

This paper introduces a flexible, versatile asset in Unity software for BCI game development to address these gaps – the P300 Dynamic Cubes (PDC) asset. The PDC asset aims to provide a dynamic, creator-friendly tool accessible to researchers, scientists and game developers that enables rapid connection and prototyping from Unity programs into BCI pipelines. Success in this project helps promote reproducibility and lowers the barrier to entry for BCI researchers interested in BCI game development. The PDC asset is available for download at <https://github.com/ekinney-lang/bci-game-jam2019>.

II. METHODS

A. P300 Stimulus Presentation and PDC Design

Stimulus presentation is designed based on the established P300 event-related visual evoked potential (VEP) [1], often used in controlling BCI visual applications. In brief, typical P300 VEP paradigms are constructed using a series of target images, pictures or letters arranged in specific patterns, like a grid, as illustrated in Figure 1 [12]. Each image is then systematically altered in a random order, changing aspects of its visual properties (e.g. contrast, background, color, etc.) to produce the P300 visual stimulus. When the object a user is attending to flashes, alterations in the brain's visual processing center can be identified via EEG, thereby allowing the BCI to infer the user's intended selection.

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Research supported by the University of Calgary Biomedical Engineering Program Funding, and the Calgary Pediatric Stroke Program.

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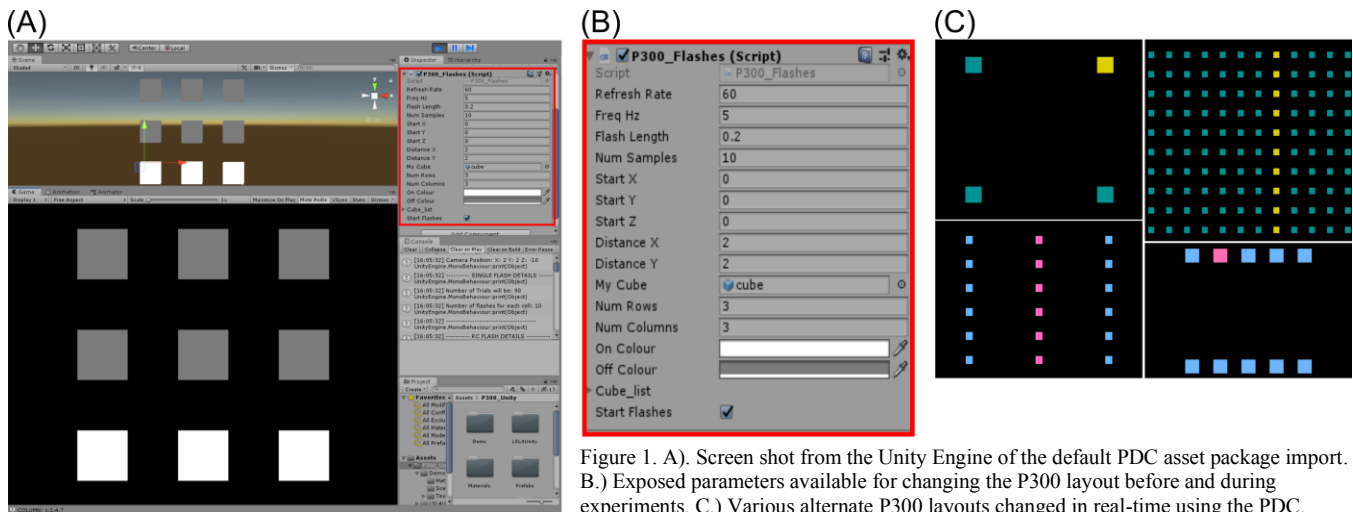


Figure 1. A.) Screen shot from the Unity Engine of the default PDC asset package import. B.) Exposed parameters available for changing the P300 layout before and during experiments. C.) Various alternate P300 layouts changed in real-time using the PDC.

To facilitate flexibility in P300 stimulus presentation, the PDC asset was designed with several key characteristics. First, critical properties required for P300 stimulus were added as a public (exposed) component on a simple geometric object within Unity (cube). Once attached, the geometric object was assigned the “prefabricated” trait within Unity. Prefabricated objects (i.e. prefabs) can be rapidly manipulated (e.g. created, destroyed, translated, etc.) while retaining a static set of properties which are shared across all associated prefabs. Through this, the PDC asset gives dynamic options for altering the exposed P300 properties by developers, allowing rapid resizing, redesign, or adaptation of the P300 stimulus to a specific set of requirements. Figure 1 (A) illustrates the in-engine view and default P300 stimulus layout of the PDC asset. Figure 1 (B) presents the readily available properties of the PDC asset. Figure 1 (C) captures several potential P300 layouts for various game designs.

B. Stimulus Marker Designation

Stimulus markers in the PDC asset were adapted from LabStreamingLayer (LSL) and the associated LSL4Unity (<https://github.com/labstreaminglayer/LSL4Unity>). Initial PDC parameters (e.g. size of P300 grid, flash frequency, number of flashes, etc.) are recorded and published as an LSL marker stream. Markers designating the P300 prefabs were given a preliminary letter code of either *s* or *r/c* depending on if the PDC stimulus flashed objects individually (*s*) or collectively as rows and columns (*r/c*). This letter code is paired with a unique numeric sequence derived from the prefabs’ position in the scene, starting from 0 for the top left and proceeding left-to-right, top-to-bottom for all *n* active P300 prefabs currently active in Unity. As an example, in a 3x3 grid of cubes individually flashing, the top-left cube would have the designation of *s,0*, while the bottom right cube would be *s,8*.

C. EEG Signal Acquisition and Processing

EEG signals were recorded from gel-based scalp electrodes using the g.tec g.GAMMASys system, g.USBamp amplifier and g.GAMMAcap headcap. A montage of eight electrodes (Fz, Cz, Pz, P3, P4, PO7, Pz, PO8) were placed in accordance with standard 10-20 system locations. A common reference electrode (FPz) was used during signal acquisition. EEG signals were recorded at a sampling rate of 256 Hz.

Signal processing and analysis were handled using custom scripts and the BCILAB toolbox [13] for MATLAB. While BCILAB was chosen as the processing pipeline here, it is important to note that the PDC asset is compatible with any BCI processing pipeline which can read in LSL markers. BCILAB was used to develop a predictive model for the BCI system, based on recorded data streamed from both the EEG and PDC asset stimulus in Unity via LSL. The BCILAB slow cortical potential signal processing paradigm script (ParadigmWindowmeans.m) was adapted for this project as the primary analysis for detecting P300 evoked potentials from EEG data. Epochs from EEG data were extracted from stimulus onset until 800ms post stimulus, based on markers sent by the PDC asset described in *Section B*. Overlapping window means were evaluated every 50ms starting from 250ms until 600ms after initial stimulus. Features of the P300 evoked response were estimated using a bandpass filter of 0.1-15 Hz on the time-series EEG for each window, with windows marked as ‘target’ or ‘non-target’ based on calibration data. Stimulus windows determined to be the highest probable ‘target’ choice sent the corresponding stimulus code back to the PDC asset for object selection in Unity.

Calibration EEG data for training the BCILAB model was recorded for preliminary proof-of-concept validation using LSL’s g.USBamp wrapper to stream EEG data, modified LSL4Unity scripts to stream Unity marker data, and recorded using LSL’s LabRecorder software. A supervised binary classification model was established using ‘target’ and ‘non-target’ classes based on when the subject actively attended to specific cube designations in the PDC asset.

New unseen EEG data for target (intended selection) and non-target (ignored selection) choices were evaluated using the obtained classifier model for real-time prototyping of the PDC asset. Proof-of-concept was evaluated for both the basic PDC asset, as well as its integration into a game environment mimicking prototypical P300 stimuli. Testing on the simple game helped verify the system’s flexibility in an active BCI environment and improve participant interest during the pilot study.

D. Experimental Setup for BCI Control

Healthy adult volunteers (*n* = 20) were recruited as part of a separate pilot study using the PDC asset as the P300 stimulus

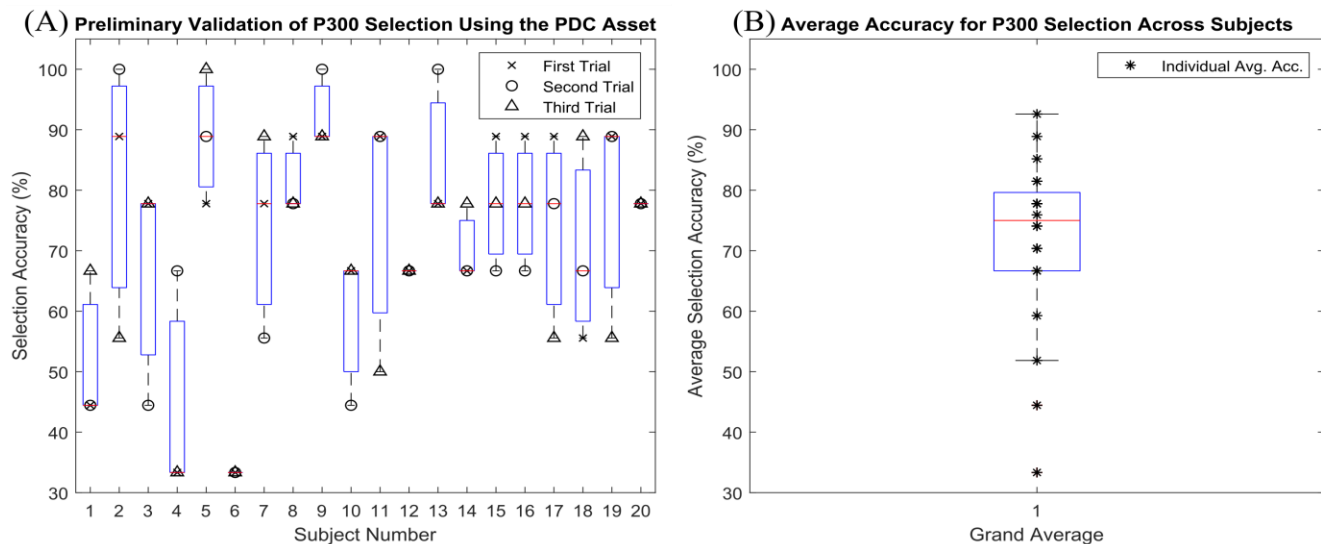


Figure 2. The left panel (A) shows individual accuracy for each of the three online BCI control trials using the PDC asset's P300 stimulus. The first 10 subjects were using the default PDC asset stimulus presentation, while subjects 11-20 used an adapted version in a game format (Figure 3, top panel). The right panel (B) shows the average online accuracy across all subjects using the PDC asset for P300 stimulus.

presentation for controlling a BCI system in Unity (submission under review for the 8th *International Brain-Computer Interface Conference*). Half the participants used the default PDC asset (Figure 2, Subjects 1-10), with the others using an adapted version incorporated into a basic game format (Figure 2, Subjects 11-20). The basic game presentation is shown in Figure 3, left panel. All participants recorded approximately 3-minutes of offline training data using the default PDC P300 presentation (Fig.1 (A)), with each cube designated as the intended target object once. The offline training data was used to initialize BCILAB's processing paradigm, with percent misclassification recorded from the predicted resubstitution loss estimate provided by BCILAB. Participants then completed 3 consecutive online BCI trials, which consisted of selecting 9 cubes at random from stimulus provided by the default or adapted PDC asset. Average online P300 accuracy was recorded based on intended versus actual selection outputs by the participant across the trials.

III. RESULTS AND OUTPUTS

A. Preliminary Validation for BCI Control

Figure 2 illustrates the accuracies recorded from the online BCI trials using the PDC asset for P300 stimulus across all participants. Misclassification rate reported by BCILAB was approximately $27.12 \pm 1.12\%$ on average over all participants. Averaging over trials and subjects gives a grand average of approximately $71.39 \pm 2.15\%$ accuracy (Median = $75 \pm 3.28\%$) for online P300 control derived from the PDC asset. The average accuracy ranged from 44.44% - 92.59% across participants, but all participants exceeded chance level for control (chance = 22.22% at $p < 0.025$ based on [14]). No statistical difference was found in average accuracy between groups using the basic and adapted PDC P300 stimulus.

B. Integration into Gaming – Outputs from the BCI Game Jam 2019

To validate how the PDC asset could be rapidly integrated into a game format, the tool was made available for the Brain-Computer Interface (BCI) Game Jam 2019. The BCI

Game Jam challenged game developers to design and create BCI-compatible games from scratch over a 48-hour competition window. The PDC asset was made available for all participants in the game jam, with live online support provided for how to include the asset into a game. A simple P300 selection game was made using the PDC asset (Fig. 3, left panel) to give developers an example demo of the tool.

Of the 9 game submissions to the BCI Game Jam, three teams incorporated and adapted the PDC asset for their game. The middle and right panels in Figure 3 presents screenshots from two of these submissions. None of the developers from the game jam had previous experience with BCI technology but were able to utilize the PDC asset for their specific game vision within the competition timeframe.

IV. DISCUSSION

Results from this project demonstrate that the PDC asset can be used to control an online P300-BCI scheme and can be successfully integrated into various game designs within Unity.

While the classification accuracy reported here may appear to be low for P300-based BCI, as many reports often cite $>90\%$ accuracy [15], the preliminary results using the default PDC stimulus still falls within a reasonable range of acceptable BCI control using the P300 compared to results [15]. This classification accuracy could potentially be improved through utilizing an alternative BCILAB processing scheme with more stringent preprocessing and artifact rejection schemes.

Though other pipelines may exist to incorporate BCI control schemes into gaming environments, the PDC asset has several benefits. First, the package is a front-end oriented system designed with non-BCI specialists in mind. The developed tool takes care of the back end BCI processing by handling all sending/receiving of data through the provided prefab object scripts. This way non-BCI experts can simply manipulate the gaming environment (e.g. Unity) without needing to adapt BCI specific scripts. Second, by exposing a limited number of critical parameters in the Unity editor,

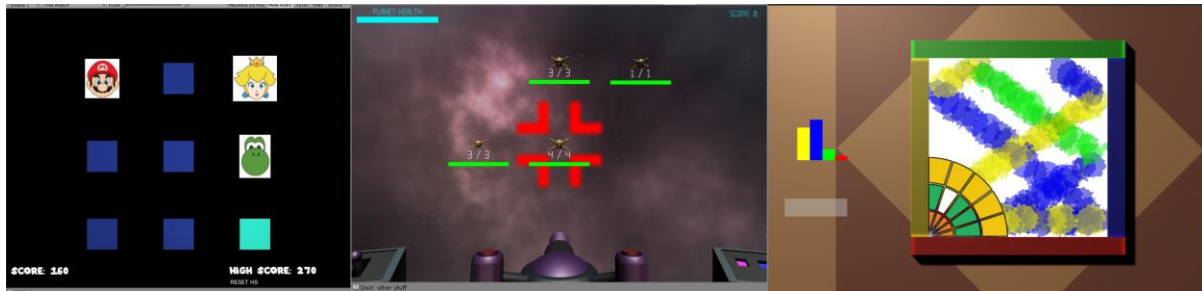


Figure 3. Screen shots showing how the PDC asset has been incorporated into various game designs. Left panel – example simple game selecting characters via P300. Middle panel – *Planet Defenders* submission to the BCI Game Jam 2019. Right Panel – *KolorKaze* submission to the BCI Game Jam 2019.

developers are able to easily alter the P300 stimulus presentation to fit their desired game design without having to concern themselves with all the specific properties required for a functional BCI system. Its inclusion by several teams during the BCI Game Jam supports that the PDC asset is intuitive enough to be utilized and extended by game developers. These results endorse the choice to focus on a front-end oriented system, accessible to a wider audience.

Future aims include making the asset more self-contained with the goal to be effectively ‘plug-and-play’ for game developers and BCI researchers alike. In addition, the authors would like to broaden the applicability of the PDC asset by providing more available BCI control paradigms (e.g. steady-state visual evoked potential SSVEP).

This study has some associated limitations. First, while the PDC asset is relatively flexible, it can only create grid based P300 patterns dynamically. Two of the exposed parameters accessible from the Unity Editor require the user to choose a specific number for “rows” and “columns”, thereby limiting alternative pattern designs. Future work could include expanding the available options to allow switching between grid patterns and other schemes (e.g. circular patterns). Additionally, the BCI processing scheme outlined in this work is done through MATLAB, outside of the Unity engine. This requires the user to launch MATLAB and run the provided initialization codes prior to starting Unity. Ongoing work is exploring how to better integrate the BCI processing pipeline directly into the Unity environment to remove the need for launching a separate program, improving the ‘plug-and-play’ capability of the PDC asset.

V. CONCLUSION

This paper outlined the P300 Dynamic Cube (PDC) asset, a dynamic and malleable P300 stimulus scheme designed for easy manipulation within the Unity Game Engine. Proof-of-concept results demonstrate the tool can effectively be integrated as the P300 stimulus for online BCI experiments using BCILAB. In addition, the PDC asset is shown to be successful as a developer friendly front-end tool, capable of implementing BCI controls into new game developments in Unity as seen in the BCI Game Jam 2019. The PDC asset therefore serves as a worthwhile and foundational addition to the available BCI game development tools.

ACKNOWLEDGMENT

The authors thank the children and families who helped

inspire the development of new BCI games and the participants of the BCI Game Jam 2019 for their submissions.

REFERENCES

- [1] L. F. Nicolas-Alonso and J. Gomez-Gil, “Brain computer interfaces, a review,” *Sensors (Basel)*, vol. 12, no. 2, pp. 1211–79, 2012.
- [2] F. Lotte *et al.*, “A review of classification algorithms for EEG-based brain-computer interfaces: A 10 year update,” *Journal of Neural Engineering*, vol. 15, no. 3. IOP Publishing, p. 031005, 01-Jun-2018.
- [3] D. A. Rohani, H. B. D. Sorensen, and S. Puthusserypady, “Brain-computer interface using P300 and virtual reality: A gaming approach for treating ADHD,” in *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC 2014*, 2014, vol. 2014, pp. 3606–3609.
- [4] B. Kerous, F. Skola, and F. Liarokapis, “EEG-based BCI and video games: a progress report,” *Virtual Real.*, 2018.
- [5] J. J. S. Norton, J. Mullins, B. E. Alitz, and T. Bretl, “The performance of 9–11-year-old children using an SSVEP-based BCI for target selection,” *J. Neural Eng.*, vol. 15, no. 5, p. 056012, Oct. 2018.
- [6] J. C. Powers, K. Bieliaieva, S. Wu, C. S. Nam, and S. D. Meriney, “The Human Factors and Ergonomics of P300-Based Brain-Computer Interfaces,” *Brain Sci. Brain Sci*, vol. 5, no. 5, pp. 318–354, 2015.
- [7] A. Myrden and T. Chau, “Effects of user mental state on EEG-BCI performance,” *Front. Hum. Neurosci.*, 2015.
- [8] M. Ahn, M. Lee, J. Choi, and S. C. Jun, “A Review of Brain-Computer Interface Games and an Opinion Survey from Researchers, Developers and Users,” *Sensors*, vol. 14, no. 8, pp. 14601–14633, 2014.
- [9] A. Vuckovic, J. A. Pineda, K. LaMarca, D. Gupta, and C. Guger, “Interaction of BCI with the underlying neurological conditions in patients: pros and cons,” *Front. Neuroeng.*, vol. 7, no. 42, pp. 1–3, Nov. 2014.
- [10] C. G. Coogan and B. He, “Brain-Computer Interface Control in a Virtual Reality Environment and Applications for the Internet of Things,” *IEEE Access*, vol. 6, pp. 10840–10849, 2018.
- [11] X. Qian *et al.*, “Brain-computer-interface-based intervention re-normalizes brain functional network topology in children with attention deficit/hyperactivity disorder,” *Transl. Psychiatry*, vol. 8, no. 1, p. 149, Dec. 2018.
- [12] L. da Silva-Sauer, L. Valero-Aguayo, A. de la Torre-Luque, R. Ron-Angevin, and S. Varona-Moya, “Concentration on performance with P300-based BCI systems: A matter of interface features,” *Appl. Ergon.*, vol. 52, pp. 325–332, 2016.
- [13] C. A. Kothe and S. Makeig, “BCILAB: A platform for brain-computer interface development,” *J. Neural Eng.*, 2013.
- [14] E. Combrisson and K. Jerbi, “Exceeding chance level by chance: The caveat of theoretical chance levels in brain signal classification and statistical assessment of decoding accuracy,” *J. Neurosci. Methods*, vol. 250, pp. 126–136, Jul. 2015.
- [15] A. Rezeika, M. Benda, P. Stawicki, F. Gembler, A. Saboor, and I. Volosyak, “Brain-computer interface spellers: A review,” *Brain Sciences*, vol. 8, no. 4. Multidisciplinary Digital Publishing Institute, p. 57, 30-Mar-2018.